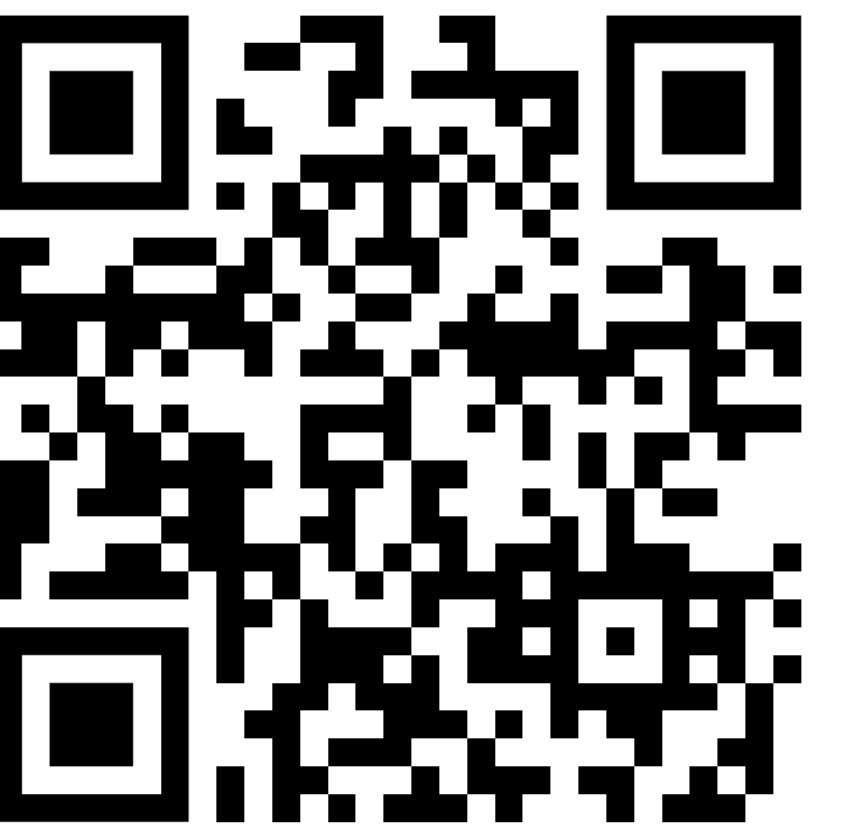


CINOC: Cardinality-Invariant Neural Operator Policies for Scalable PDE Control

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Introduction

The Challenge: Controlling partial differential equations (PDEs) is critical, but existing learning-based multi-agent policies generally rely on fixed-dimensional representations.

Our Solution: We recast PDE policy synthesis as an operator learning problem, mapping state fields to continuous control functions.

The Result: End-to-end training via differentiable physics yields policies with cardinality invariance, allowing zero-shot transfer to significantly larger agent populations.

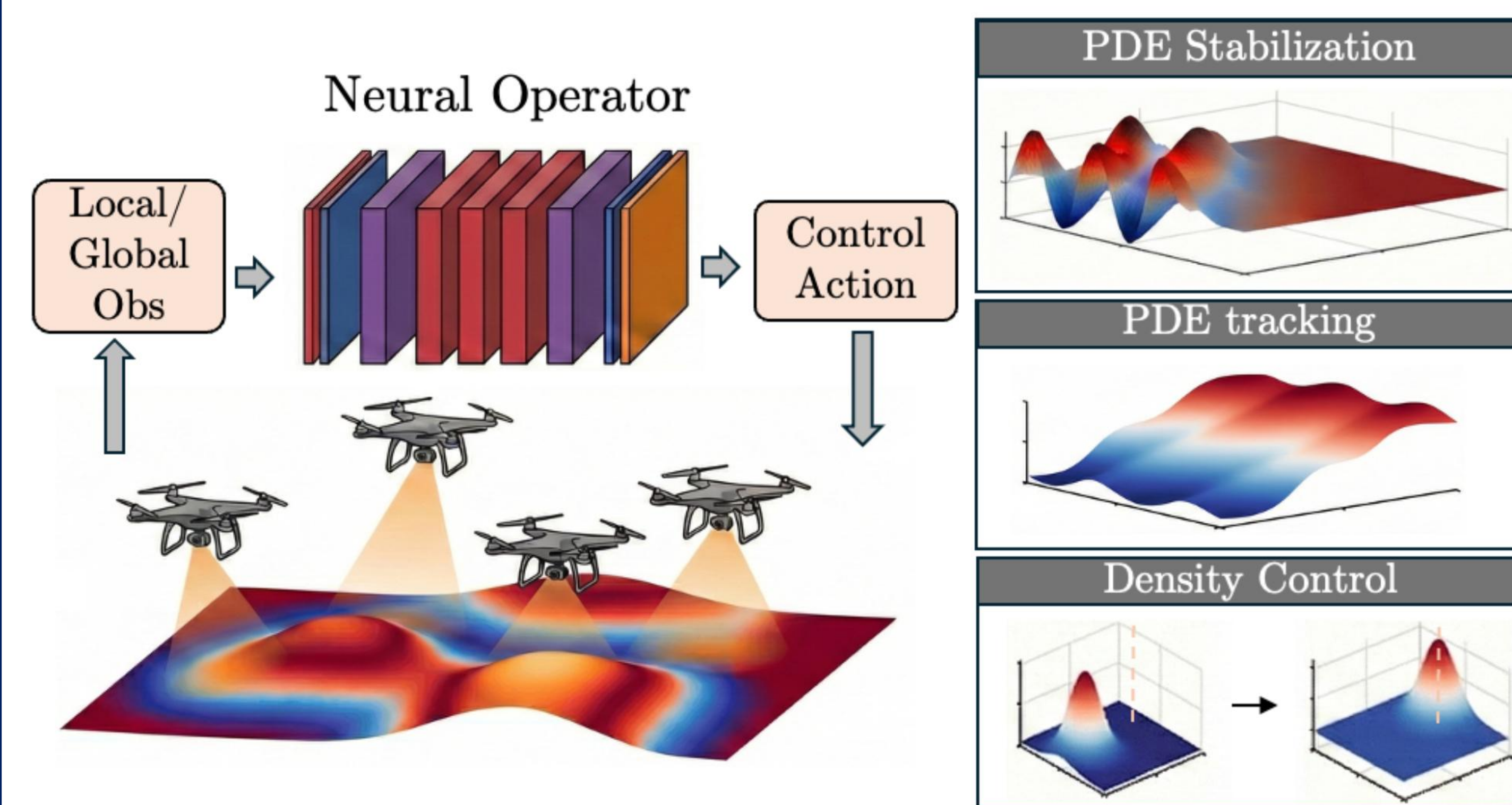


Figure 1: A shared neural operator maps PDE field observations to control actions for multiple agents, enabling zero-shot scalability across agent populations. Framework handles diverse tasks: (top right) chaos stabilization, (middle right) reference tracking, and (bottom right) density transport.

Contributions

- **Neural Operator Framework:** Control policies parameterized as neural operators yield configuration-invariant behavior across variable sensing/actuation/population setups.
- **Differentiable-Physics Optimization:** Policies trained end-to-end by differentiating through PDE solvers, providing scalable and efficient policy gradients.
- **Cardinality-Invariant Multi-Agent Control:** An emergent self-normalization phenomenon enables consistent multi-agent behavior independent of agent count.
- **Mean-Field Grounding:** Finite-population policy gradient learning is formally linked to a mean-field limit, theoretically establishing cardinality invariance.
- **Diverse PDE Benchmarks:** Demonstrated across stabilization, tracking, and density transport on nonlinear, diffusion, and chaotic spatiotemporal dynamics.

Problem formulation and Methodology

$$\min_{\mathbf{u}^M, \mathbf{v}^M} \mathcal{J} = \mathbb{E}_{\psi \sim \mathcal{D}_{\text{prob}}} \left[\int_0^T \ell(z(\mathbf{x}, t), \mathbf{u}^M(t), \mathbf{v}^M(t), \psi) dt \right]$$

$$\text{s.t. } \frac{\partial z}{\partial t} = \mathcal{F}(z(\mathbf{x}, t); \psi) + \sum_{i=1}^M \mathcal{B}(\mathbf{x}, \mathbf{u}_i(t), \xi_i(t)),$$

$$\dot{\xi}_i(t) = \mathbf{v}_i(t), \quad i = 1, \dots, M,$$

$$\mathcal{C}(z(\mathbf{x}, t), \mathbf{u}^M(t), \mathbf{v}^M(t), \xi^M(t), \psi) \leq 0.$$

This formulation defines the objective for our framework:

- **Orange (Continuous Control Policy):** the variables we are actively optimizing. The neural operator maps local observations to the control functions $\mathbf{u}^M$ and velocities $\mathbf{v}^M$.
- **Purple (System Dynamics):** represents the PDE state field (z).
- **Teal (Decentralized Actuation):** The spatial coordinates (ξ) define the position of each agent, whose effect on the field is given by $\mathcal{B}$.
- **Gray (Distributional Robustness):** we evaluate the expected cost over a distribution of parameters (ψ , e.g. initial and reference shapes).

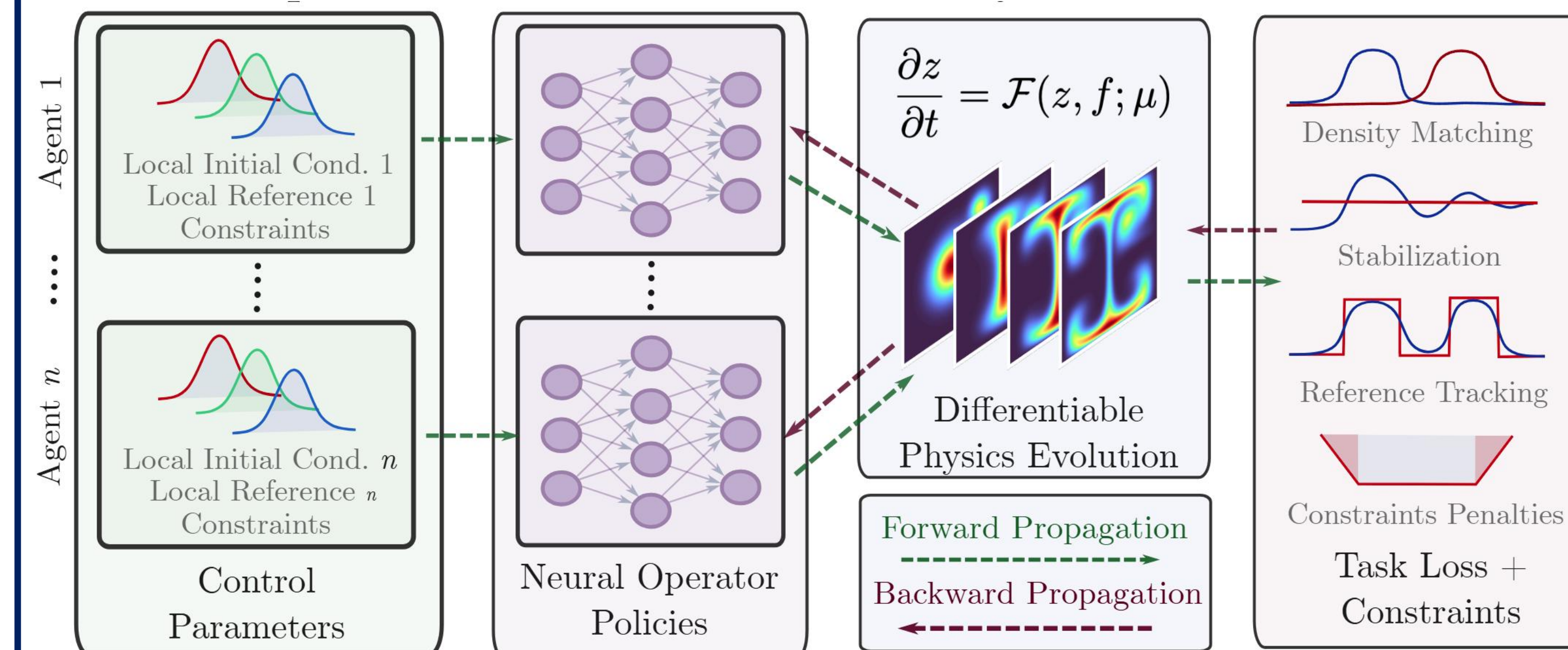


Figure 2: Schematic of the multi-agent operator learning for PDE control. The PDE solver acts as a differentiable layer. The operator policy (shared among the agents) takes error observations and outputs control actions. Gradients propagate through the solver to update

Theoretical Analysis

- **Mean-Field Gradient Consistency :** As $M \rightarrow \infty$, discrete policy gradients $\nabla_{\theta} \mathcal{J}_M$ converge to the continuous mean-field limit $\mathcal{D}_{\theta} \mathcal{J}_{\infty}$ (Thm. 1), guaranteeing the learned operator captures true PDE dynamics rather than swarm-size artifacts.
- **Emergent Self-Normalization:** Effort regularization implicitly drives each agent toward $u_i \approx \mathcal{O}(1/M)$, keeping aggregate forcing bounded without any explicit communication of swarm size M , directly producing cardinality-invariant behavior.

Results

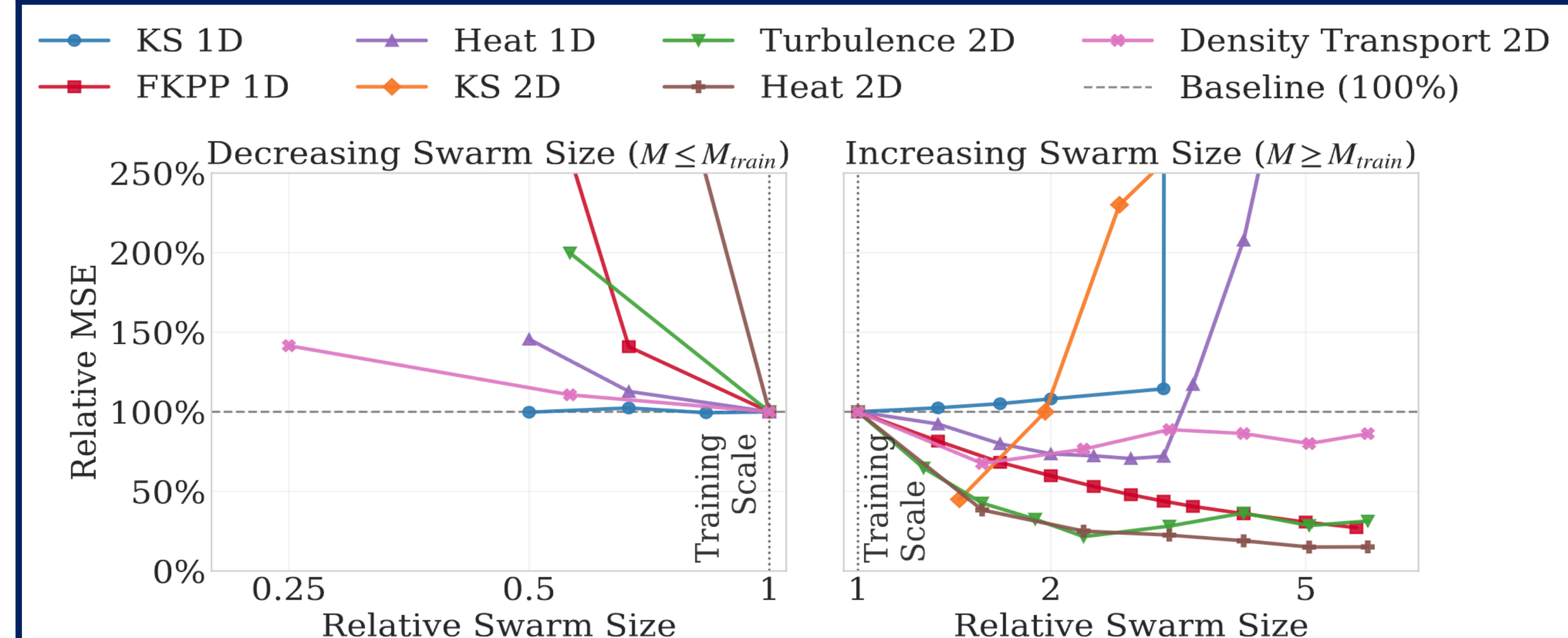


Figure 3: Relative MSE vs. swarm size normalized to training scale M_{train} (dashed = baseline). Left: scale-down ($M \leq M_{\text{train}}$); Right: scale-up ($M \geq M_{\text{train}}$). 2D environments (Density 2D, Turbulence 2D) show broader scaling margins due to higher geometric capacity; chaotic regimes (KS equation) show tightest margins as forcing errors amplify rapidly.

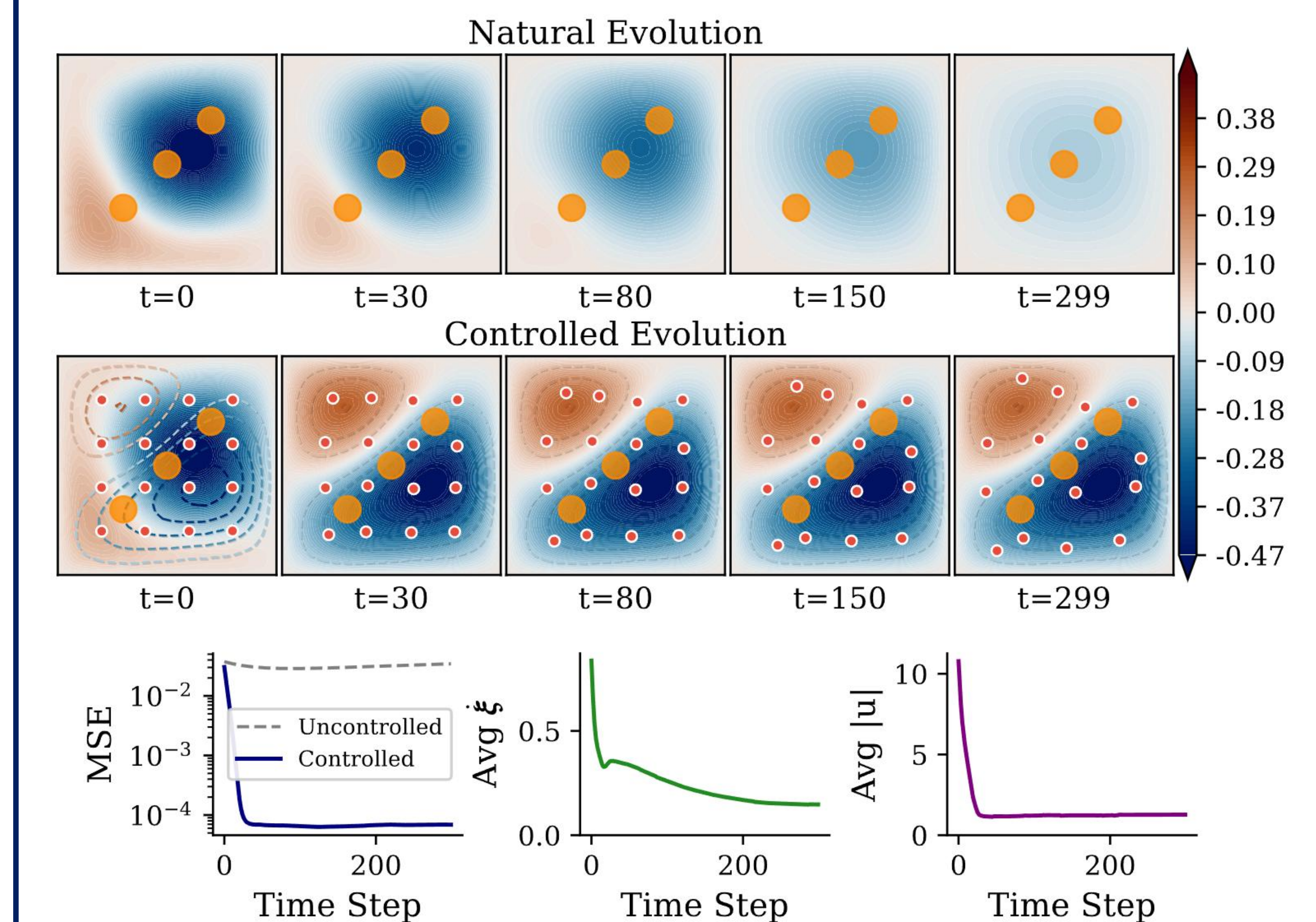


Figure 4: Reference tracking of 2D Heat equation. Top: Natural (uncontrolled) evolution of $tu(x, t)$ showing diffusion. Middle: Controlled evolution of the thermal field $u(x, t)$ where agents (red dots) regulate the temperature field toward dashed reference contours while avoiding static obstacles (orange circles). Bottom: Tracking MSE, average agent velocity (ξ) and average agent control intensity ($|u|$).

Conclusion

CINOC reformulates PDE control as operator learning, achieving zero-shot cardinality invariance across agent populations. Emergent self-normalization, grounded by mean-field gradient consistency, enables bounded aggregate forcing without explicit swarm-size communication. Future work targets parametric operators conditioned on physical coefficients and scaling to high-dimensional industrial environments.

